

$$\begin{cases} X_{24} = X_{14} - X_{13} \\ X_{2j} = X_{1j} - X_{13} & j=5, \dots, n-2 \\ X_{2n-1} = X_{1n-1} - X_{13} \\ X_{2n} = -X_{13} \end{cases}$$

dots highlighted

$$+0 \text{ proj:} \rightarrow \begin{cases} X_{2j} = X_{1j} - X_{13} & j=4, \dots, n-2 \\ X_{2n-1} = -X_{13} \end{cases} \quad (+0)$$

$$\begin{aligned} -1 \text{ proj:} & \begin{cases} X_{13} = X_{n3} - X_{n2} \\ X_{1j} = X_{nj} - X_{n2} & j=4, \dots, n-3 \\ X_{1n-2} = X_{nn-2} - X_{n2} \\ X_{1n-1} = -X_{n2} \end{cases} \xrightarrow{X_{1n-1}=0} \begin{cases} X_{13} = X_{n3} \\ X_{1j} = X_{nj} \\ X_{1n-2} = X_{nn-2} \\ X_{n2} = 0 \end{cases} \quad \begin{matrix} \times \\ (0\text{-zero, i.e. no quotient} \\ \text{on } 1, \dots, n-1) \end{matrix} \end{aligned}$$

$$\begin{aligned} -2 \text{ proj:} & \begin{cases} X_{n2} = X_{n-12} - X_{n-11} \\ X_{nj} = X_{n-1j} - X_{n-11} & (j=3, \dots, n-4) \\ X_{nn-3} = X_{n-1n-3} - X_{n-11} \\ X_{nn-2} = -X_{n-11} \end{cases} \xrightarrow{X_{nn-1}=0} \begin{cases} X_{n2} = X_{n-12} \\ X_{nj} = X_{n-1j} \\ X_{nn-3} = X_{n-1n-3} \\ X_{nn-2} = 0 \end{cases} \quad \begin{matrix} \times \\ (0\text{-zero, i.e. no quotient} \\ \text{on } 1, \dots, n-1) \end{matrix} \end{aligned}$$

$$\begin{aligned} -3 \text{ proj:} & \begin{cases} X_{n-1,1} = X_{n-2,1} - X_{n-2,n} \\ X_{n-1,j} = X_{n-2,j} - X_{n-2,n} & (j=2, \dots, n-5) \\ X_{n-1,n-4} = X_{n-2,n-4} - X_{n-2,n} \\ X_{n-1,n-3} = -X_{n-2,n} \end{cases} \xrightarrow{X_{1n-1}=0} \begin{cases} X_{n-1,j} = X_{n-2,j} - X_{n-2,1} & j=(2, n-4) \\ X_{n-1,n-3} = -X_{n-2,1} \end{cases} \quad \begin{matrix} (-2) \equiv + (n-1-2) \end{matrix} \end{aligned}$$

$$\begin{aligned} +1 \text{ proj:} & \begin{cases} X_{3j} = X_{2j} - X_{2n} & (j=5, \dots, n) \\ X_{31} = -X_{2n} \end{cases} \xrightarrow{(+1)} \begin{cases} X_{3j} = X_{2j} - X_{2n} & (j=5, \dots, n-1) \\ X_{31} = -X_{2n} \end{cases} \end{aligned}$$

$$\begin{aligned} +2 \text{ proj:} & \begin{cases} X_{4j} = X_{3j} - X_{35} & (j=6, \dots, n) \\ X_{41} = X_{31} - X_{35} \\ X_{42} = -X_{35} \end{cases} \xrightarrow{(+2)} \begin{cases} X_{4j} = X_{3j} - X_{35} & (j=7, \dots, n-1) \\ X_{41} = X_{31} - X_{35} \\ X_{42} = -X_{35} \end{cases} \end{aligned}$$

$$\begin{aligned} +3 \text{ proj:} & \begin{cases} X_{5j} = X_{4j} - X_{46} & (j=7, \dots, n) \\ X_{5i} = X_{4i} - X_{46} & (i=1, 2) \\ X_{53} = -X_{46} \end{cases} \xrightarrow{(+3)} \begin{cases} X_{5j} = X_{4j} - X_{46} & (j=7, \dots, n-1) \\ X_{5i} = X_{4i} - X_{46} & (i=1, 2) \\ X_{53} = -X_{46} \end{cases} \end{aligned}$$

in general, $\textcircled{+l}$ with $l \in \{1, \dots, n-4\}$

$$X_{2+l,j} = X_{1+l,j} - X_{1+l,3+l} \quad j \in \{4+l, \dots, n\}$$

$$X_{2+l,i} = X_{1+l,i} - X_{1+l,3+l} \quad i \in \{1, l-1\}$$

$$X_{2+l,l} = -X_{1+l,3+l}$$

$$X_{2+l,j} = X_{1+l,j} - X_{1+l,3+l} \quad (j=4+l, \dots, n-1)$$

$$X_{2+l,j} = X_{1+l,j} - X_{1+l,3+l} \quad (j = 1, \dots, l-1)$$

$$X_{2+l, l} = -X_{1+l, 3+l}$$

 $(+l)$ for $n-1$

we can get $+0, +1, +2, \dots, +(n-4), + (n-3)$ at the $n-1$ level by projecting. we only

missig + (n-2) shift (downstairs)

using

$$\begin{cases} X_{1,j} = X_{n-1,j} - X_{n-1,2} & j = 3, \dots, n-2 \\ X_{1,n-2} = -X_{n-1,2} \end{cases}$$

$$X_{1n-2} = -X_{n-1,2}$$

57	51	52	53	54
	61	62	63	64
		72	73	74
			13	14
				24

$$\begin{array}{rcl} 62 & 63 & 64 \end{array} \quad \begin{array}{rcl} 6 & n-1 & \\ 13 & 14 & \end{array} \quad \begin{array}{rcl} 1 & n-1 & \\ 24 & 2 & n-1 \\ & 1 & \\ & n-3, 4-1 & \end{array}$$

when plotting 2-zeros from $n=7$ we are missing

51	52	53	54
	62	63	64
		13	14
			24

and

62	63	64	65
	13	14	15
		24	25
			35

let's look at a 3-zero for $n = 7$

$$X_{46} = X_{16} - X_{15}$$

$$X_{46} = X_{26} - X_{25}$$

$$X_{46} = X_{36} - X_{35}$$

$$X_{47} = \quad - \quad X_{15}$$

$$X_{47} = X_{27} - X_{25}$$

$$X_{47} = X_{37} - X_{35}$$

13	14	15	16	17
24	25	26	27	
	35	36	37	
		46	47	
			57	

f1	d1	z1	u1	z1
f5	ds	zs	us	
f3	ds	zs		
f11	du			
f2				

Conjecture :

gory	fian	$7 \rightarrow 6$	ne	re but	all	1-zero		
gory	fian	$8 \rightarrow 7$	ne	re but	all	1-zero	and	2-zero
gory	fian	$9 \rightarrow 8$	ne	re but	all	1-zero	and	2-zero

going from $1n \rightarrow 9$ we recover all 1-zeros and 2-zeros at 3-zeros

let's start by showing we recover all the 1-zero. we get $+0, \dots, +n-3$ from the 1-zeros, as shown.

To get the $+n-2$ one, let's start from the $+n-2$ 2-zero for n

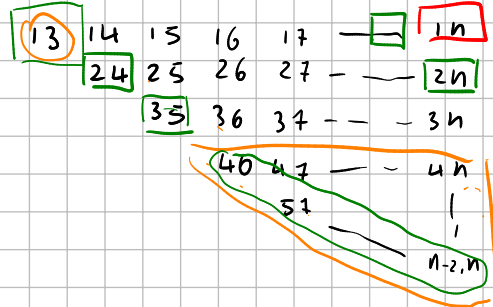
$$X_{35} = X_{15} - X_{14}$$

$$X_{35} = X_{25} - X_{24}$$

$$\left. \begin{aligned} X_{3i} &= X_{1i} - X_{14} \\ X_{3i} &= X_{2i} - X_{24} \end{aligned} \right\} i = 6, \dots, n-2$$

$$X_{3n-1} = X_{1n-1} - X_{14}$$

$$X_{3n-1} = X_{2n-1} - X_{24}$$

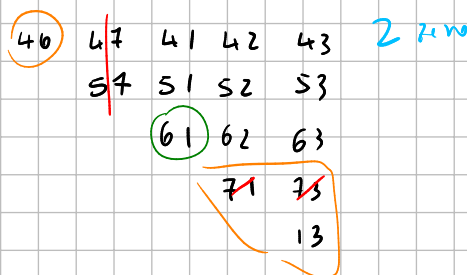
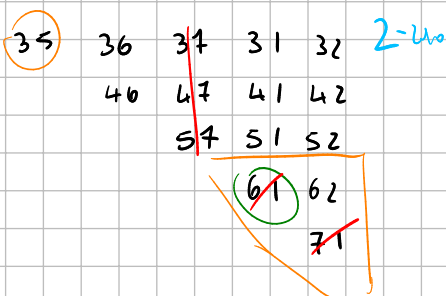
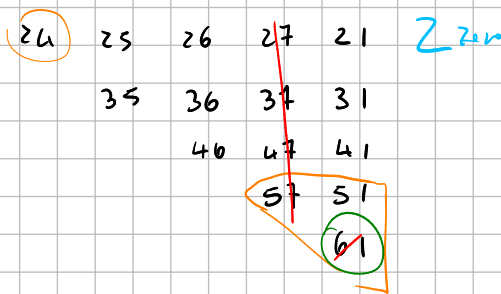
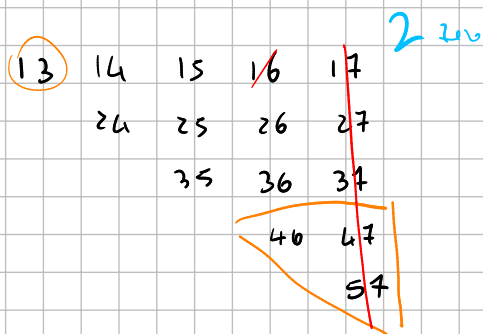


cases

Conjecture

- shift so $1n-1$ is in the bottom BF $\rightarrow$ 2-zero
- shift so $1n-1$ is in the top BF $\rightarrow$ 1-zero
- shift so $1n-1$ gets rid of the 24 or 35 spots $\rightarrow$ 1-zero
- shift so $1n-1$ gets in the n

13 14 15 16 17



~~54~~ 51 52 53 54 1-zero
 61 62 63 64
~~71~~ 73 74
 13 14
 24

61 62 63 64 65 1-zero
~~71~~ 73 74 75
 13 14 15
 24 25
 35

-2

~~72~~ 73 74 75 76 1-zero
 13 14 15 16
 24 25 26
 35 36
 46

look at the $(+n-2)$ 2-zero

~~n-1,1~~ n-1,2 n-1,3 n-1,4 ... n-1,n-2
~~n-2~~ ~~n-3~~ ~~n-4~~ ... ~~n,n-2~~
 1 3 1 4 ... 1, n-1
 2 4 ... 2, n-2
 ...
 n-4, n-2

→ this is $K_1(+n-2)$ 1-zero

→ we get all the 1-zeros

expectation: the $+n-2, +n-3, +n-4$ 3-zeros for $n=8$ project to
 the $+n-2, +n-3, +n-4$ 2-zeros for $n=7$

look at this $n=7$ 3-zero

~~54~~ 51 52 53 54
~~61~~ 62 63 64
~~72~~ 73 74
 13 14
 24

-3 3-zero

~~$X_{15} = X_{13} - X_{72}$~~
 $X_{13} = X_{03} - X_{02}$
 $X_{13} = X_{53} - X_{52}$
 ~~$X_{14} = X_{24} - X_{72}$~~
 $X_{14} = X_{04} - X_{62}$
 $X_{14} = -X_{52}$

$X_{16} = 0$
 →

$X_{13} = X_{03} - X_{02}$
 $X_{13} = X_{53} - X_{52}$
 $X_{14} = X_{04} - X_{62}$
 $X_{14} = -X_{52}$

→ we can linearly get all the zeros

$$X_{46} = X_{16} - X_{15}$$

$$X_{46} = X_{26} - X_{25}$$

$$X_{46} = X_{36} - X_{35}$$

$$X_{47} = \quad - X_{15}$$

$$X_{47} = X_{27} - X_{25}$$

$$X_{47} = X_{37} - X_{35}$$

13	14	15	16	17
24	25	26	27	
	35	36	37	
		46	47	
			57	



$$X_{15} = \quad - X_{47}$$

$$X_{46} = X_{16} + X_{47}$$

$$X_{46} = X_{26} - X_{25}$$

$$X_{46} = X_{36} - X_{35}$$

$$X_{15} = \quad - X_{47}$$

$$X_{47} = X_{27} - X_{25}$$

$$X_{47} = X_{37} - X_{35}$$

13	14	15	16	17
24	25	26	27	
	35	36	37	
		46	47	
			57	

27	17	13	11	31
47	37	35	25	21
	46	45	43	41
		57	55	53
			67	63

46	47	41	42	43
	57	51	52	53
		61	62	63
		72	73	
			13	

$n=5$

13	14	15
24	25	
	35	

→

$$\begin{cases} X_{35} = X_{25} - X_{24} \\ X_{35} = \quad - X_{14} \end{cases} \rightarrow$$

$$-X_{14} = X_{25} - X_{24} \rightarrow$$

$$X_{14} = -X_{35}$$

$$\begin{cases} X_{25} = X_{24} - X_{14} \\ X_{14} = -X_{35} \end{cases}$$

we shift $X_{14} = -X_{35}$ into $X_{25} = -X_{13}$?

$$\begin{array}{l} +1 \quad 25 = -41 \\ +2 \quad 31 = -52 \end{array}$$



$$\begin{cases} X_{25} = X_{24} - X_{14} \\ X_{14} = -X_{35} \end{cases} \xrightarrow{+2} \begin{cases} X_{42} = X_{41} - X_{31} \\ X_{31} = -X_{52} \end{cases}$$

$$\begin{cases} X_{35} = X_{25} - X_{24} \\ X_{35} = \quad - X_{14} \end{cases} \xrightarrow{+2} \begin{cases} X_{52} = X_{42} - X_{41} \\ X_{52} = -X_{31} \end{cases}$$

$n=6$

51 52 53 54
62 63 64
13 14
24

special:

↑
equivalent

$$X_{24} = -X_{53}$$

$$X_{24} = -X_{53}$$

$$X_{24} = X_{64} - X_{63}$$

$$\Leftrightarrow -X_{53} = X_{64} - X_{63} \Leftrightarrow$$

$$X_{46} = X_{36} - X_{35}$$

$$X_{41} = X_{31} - X_{35}$$

$$X_{24} = X_{14} - X_{13}$$

$$-X_{53} = X_{14} - X_{13}$$

$$X_{42} = -X_{35}$$

(+2)

35 36 31 32
46 41 42

51 52
62

start with (+4) 1-zero

51 52 53 54
62 63 64
13 14
24

→

$$\begin{cases} X_{62} = X_{52} - X_{51} \\ X_{63} = X_{53} - X_{51} \\ X_{64} = -X_{51} \end{cases}$$

equivalent to

13 14 15 16
24 25 26
35 36
40

3-zero

→

$$\begin{cases} X_{46} = X_{36} - X_{35} \\ X_{46} = X_{26} - X_{25} \\ X_{46} = -X_{15} \end{cases}$$

projection:

51 52 53 54
~~62 63 64~~
13 14
24

→

52 53 54
13 14
24

no constraint

13 14 15 16
24 25 26
35 36
40

→

13 14 15
24 25
35

no constraint

57 51 52 53 54
61 62 63 64
72 73 74
13 14
24

Proj
→

51 52 53 54
62 63 64
13 14
24

57 51 52 53 54
61 62 63 64
72 73 74
13 14
24

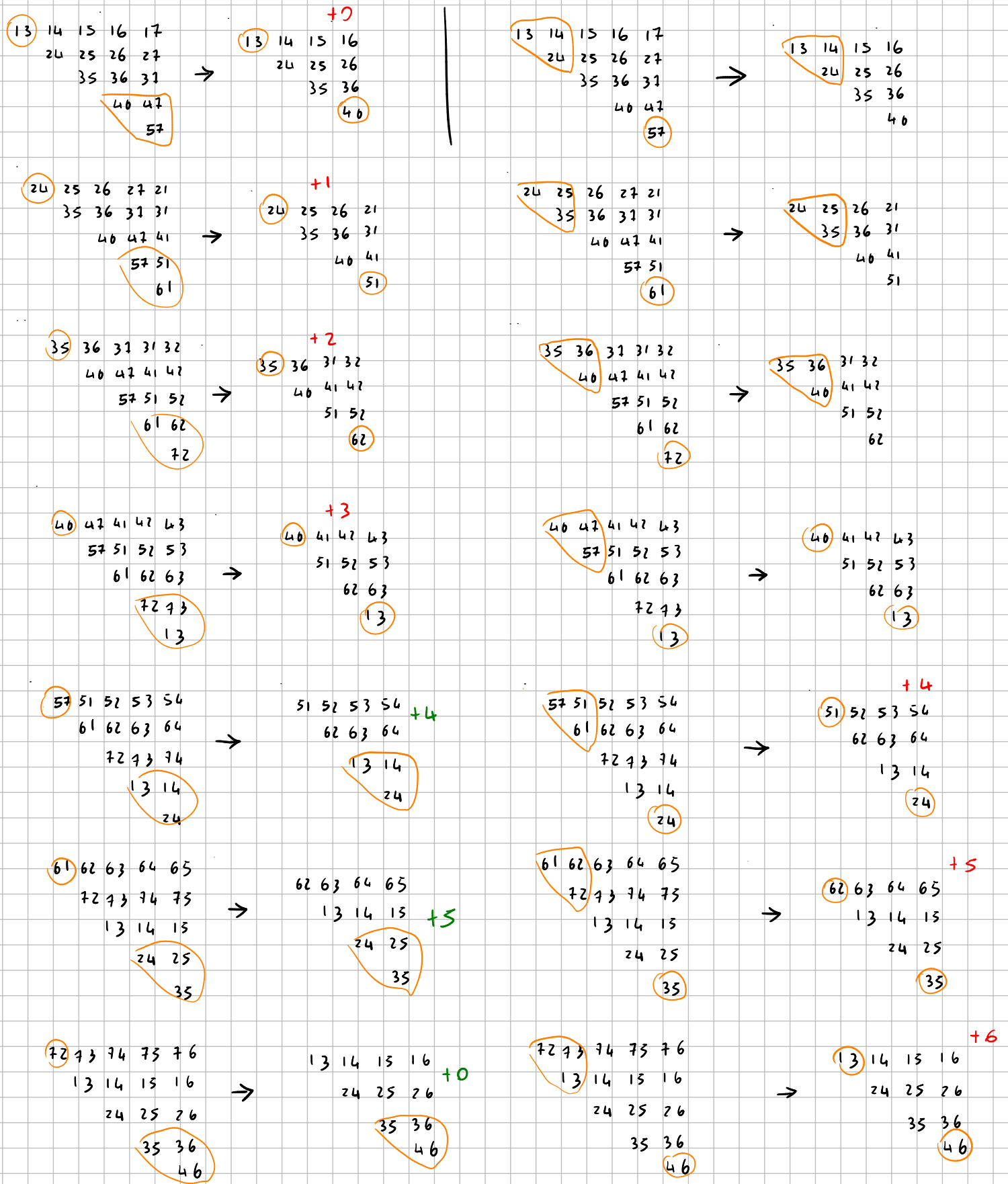
Proj
→

51 52 53 54
62 63 64
13 14
24

Conclusion:

- 4 of the 2-zeros project to 2-zeros
- 3 of the 2-zeros project to 1-zero

somehow we recover the remaining 2-zeros (2 of them)
 from the 3-zeros, but they are equivalent to 2-zeros



13 14 15 16 17
24 25 26 27
35 36 37
40 41
52

→

13 14 15 16
24 25 26
35 36
40 +0

24 25 26 27 28
35 36 37 38
40 41 42
52 53
64

→

24 25 26 27
35 36 37 +1
40 41
52

35 36 37 38 39
40 41 42 43
52 53 54
64 65
76

→

35 36 37 38
40 41 42 +2
52 53
64

40 41 42 43 44
52 53 54 55
64 65 66
76 77
88

→

40 41 42 43
52 53 54 +3
64 65
76

52 53 54 55 56
64 65 66 67
76 77 78
88 89
90

→

52 53 54 55
64 65 66 +4
76 77
88

64 65 66 67 68
76 77 78 79
88 89 90
91 92
93

→

64 65 66 67
88 89 90
91 92
93 X

76 77 78 79 80
88 89 90 91
92 93 94
95 96
97

→

13 14 15 16
24 25 26
35 36
40 X

current algorithm: we can get everything

$$\begin{array}{cccc} 24 & 25 & 26 & 27 & 21 \\ & 35 & 36 & 37 & 31 \\ & & 46 & 47 & 41 \\ & & & 57 & 51 \\ & & & & 61 \end{array}$$

↕ equivalent

$$\begin{array}{cccc} 57 & 51 & 52 & 53 & 54 \\ 61 & 62 & 63 & 64 & \\ 72 & 73 & 74 & & \\ 13 & 14 & & & \\ & 24 & & & \end{array}$$

$$\begin{array}{cccc} & & +1 & \\ 24 & 25 & 26 & 21 \\ & 35 & 36 & 31 \\ & & 46 & 41 \\ & & & 51 \end{array}$$

↕ equivalent

$$\begin{array}{cccc} & & +4 & \\ 51 & 52 & 53 & 54 \\ & 62 & 63 & 64 \\ & 13 & 14 & \\ & & 24 & \end{array}$$

$$\begin{cases} X_{46} = X_{36} - X_{35} \\ X_{46} = X_{26} - X_{25} \\ X_{41} = X_{31} - X_{35} \\ X_{41} = -X_{25} \end{cases}$$

$$\begin{cases} X_{13} = X_{63} - X_{62} \\ X_{13} = X_{53} - X_{54} \\ X_{14} = X_{64} - X_{62} \\ X_{14} = -X_{62} \end{cases}$$

$$X_{13} = X_{63} - X_{62}$$

$$X_{13} = X_{53} - X_{52}$$

$$X_{46} = X_{62} - X_{25}$$

$$X_{14} = -X_{62}$$

$$X_{13} = X_{63} - X_{46} + X_{14}$$

$$X_{13} = X_{53} + X_{14}$$

$$X_{46} = X_{62} - X_{25}$$

$$X_{14} = -X_{62}$$

$$\begin{cases} X_{46} = X_{63} - X_{53} \\ X_{14} = X_{13} - X_{35} \\ X_{46} = X_{62} - X_{25} \\ X_{14} = -X_{62} \end{cases}$$

→ not all 2-zeros are independent! (n=7)

Focus going forward: prove then are no crossings with a short.